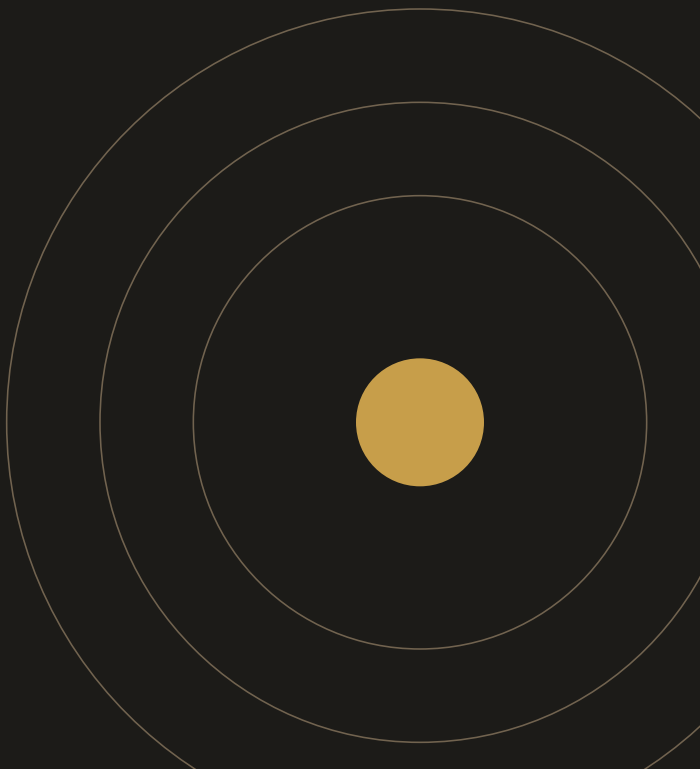
BUILD MANUAL

Sisyphus Desk Display

Complete hardware assembly, wiring, software setup, commissioning, and maintenance instructions for the ESP32 animated weather display.

CONTROLLER	DISPLAY	FIRMWARE
ESP32-WROOM-32	1.54 in ST7789 240 x 240 TFT	Arduino IDE

A tiny world of weather, moonlight, persistence, and one very patient mallard.



SECTION 00

Before you begin

This manual is written for a first-time builder using the specific ESP32 and ST7789 hardware selected for this project. Read the complete wiring and power sections before connecting USB power.

What this device does

The finished desk display shows local time, temperature, current weather, the active moon phase, a continuously animated Sisyphus scene, occasional mallard visits, and a persistent mileage counter. The ESP32 connects to Wi-Fi for time and weather data. The display animation continues locally even if the internet connection is temporarily unavailable.

How to use this manual

Section	Purpose
01 - Parts and tools	Confirm that every required component is available.
02 - Wiring	Connect the TFT and optional light sensor safely.
03 - Software	Install Arduino IDE support and required libraries.
04 - Firmware	Configure, compile, and upload the supplied sketch.
05 - First start	Join Wi-Fi and verify live data.
06 - Enclosure	Move the tested circuit into its finished housing.
07 - Testing	Run a complete commissioning checklist.
08 - Troubleshooting	Diagnose display, USB, Wi-Fi, and data problems.
09 - Maintenance	Protect stored mileage and keep the device reliable.

POWER AND LOGIC WARNING

Use 3.3 V logic for the display connections. Never connect an ESP32 GPIO pin directly to 5 V. Disconnect USB power before moving wires. A direct BLK-to-GPIO connection is permitted only when the display module documentation confirms that BLK is a low-current control input. Otherwise, connect BLK to 3.3 V for full brightness or use the transistor circuit described later in this manual.

SECTION 01

Parts and tools

Use the exact display driver and pin labels shown here. Similar-looking TFT modules may use a different controller or pin order.

Required electronics

Qty	Part	Notes
1	ESP32 ESP-32S development board	ESP-WROOM-32 module, USB-C connector, CP2102 USB interface.
1	1.54 inch ST7789 TFT module	240 x 240 IPS display with eight pins: GND, VCC, SCL, SDA, RES, DC, CS, BLK.
1	USB-C data cable	Must carry both power and data. Charge-only cables will not upload firmware.
1	5 V USB power supply	Use a reputable supply rated for at least 1 A.
8+	Female-to-male or female-to-female jumpers	Choose connectors that match the installed headers.
1	Solderless breadboard	Recommended for initial testing before permanent assembly.
1	Enclosure and mounting hardware	Install only after the complete circuit passes testing.

Optional electronics

Qty	Part	Purpose
1	BH1750 / GY-302 light sensor	Measures room brightness for automatic dimming.
1	Logic-level N-channel MOSFET or NPN transistor	Safely controls a backlight that draws more current than a GPIO pin can supply.
1	1 k-ohm resistor	Gate/base series resistor for the backlight control circuit.
1	10 k-ohm resistor	MOSFET gate pull-down when a MOSFET backlight circuit is used.
1	Perfboard and headers	Creates a more durable final assembly.

Tools

Small screwdriver, flush cutters, wire stripper, soldering iron and solder for permanent assembly, multimeter, computer running Windows/macOS/Linux, and access to a 2.4 GHz Wi-Fi network.

No series resistors are normally required on the SPI signal lines when both devices use 3.3 V logic. The optional backlight circuit is the exception; its parts depend on the electrical design of the particular TFT module.

SECTION 02

System layout

Complete and test the project on a breadboard before installing it in an enclosure. This keeps every connection visible while faults are easy to correct.

Connection groups

Group	ESP32 resources	Function
TFT data	GPIO 18, 23, 17, 16, 5	SPI clock/data plus display reset, data-command, and chip select.
TFT power	3.3 V and GND	Powers the display logic and establishes a common ground.
Backlight	GPIO 25 or 3.3 V	PWM dimming through a safe control input/circuit, or constant full brightness.
Optional light sensor	GPIO 21 and 22	I2C data and clock for BH1750 ambient-light readings.
USB	USB-C connector	Firmware upload and normal 5 V input power.

The ESP32 uses 3.3 V logic even though the development board receives 5 V through USB. All modules must share the same GND connection. A missing common ground can produce a blank display, corrupted graphics, or unreliable sensor readings.

Orientation note

The display can be mounted in any 90-degree orientation. The final firmware rotation setting must match the physical enclosure. Do not cross wires merely to rotate the image; rotation is handled in software.

SECTION 02A

Wire the ST7789 display

Do not rely on the physical order of pins in a photograph. Read the text printed beside each pin on your module and connect by label.

TFT label	Meaning	ESP32 connection	Typical board marking
GND	Ground	GND	GND
VCC	Display power	3.3 V	3V3
SCL / SCK	SPI clock	GPIO 18	D18
SDA / MOSI	SPI data from ESP32	GPIO 23	D23
RES / RST	Display reset	GPIO 17	D17 / TX2
DC	Data or command select	GPIO 16	D16 / RX2
CS	Chip select	GPIO 5	D5
BLK / LED	Backlight	See backlight section	GPIO 25 or 3.3 V

Pin-name translation

On this display, SDA means the SPI data input and connects to MOSI. It is not an I2C SDA connection. SCL is the SPI clock and connects to SCK. The ESP32 board commonly prints these connections as D23 and D18. In Arduino code, use the GPIO numbers 23 and 18.

SECTION 02A

Wiring diagram

This hand-drawn diagram records the physical connections used for the working display. Follow the written pin table on the preceding page whenever a wire color is difficult to distinguish.

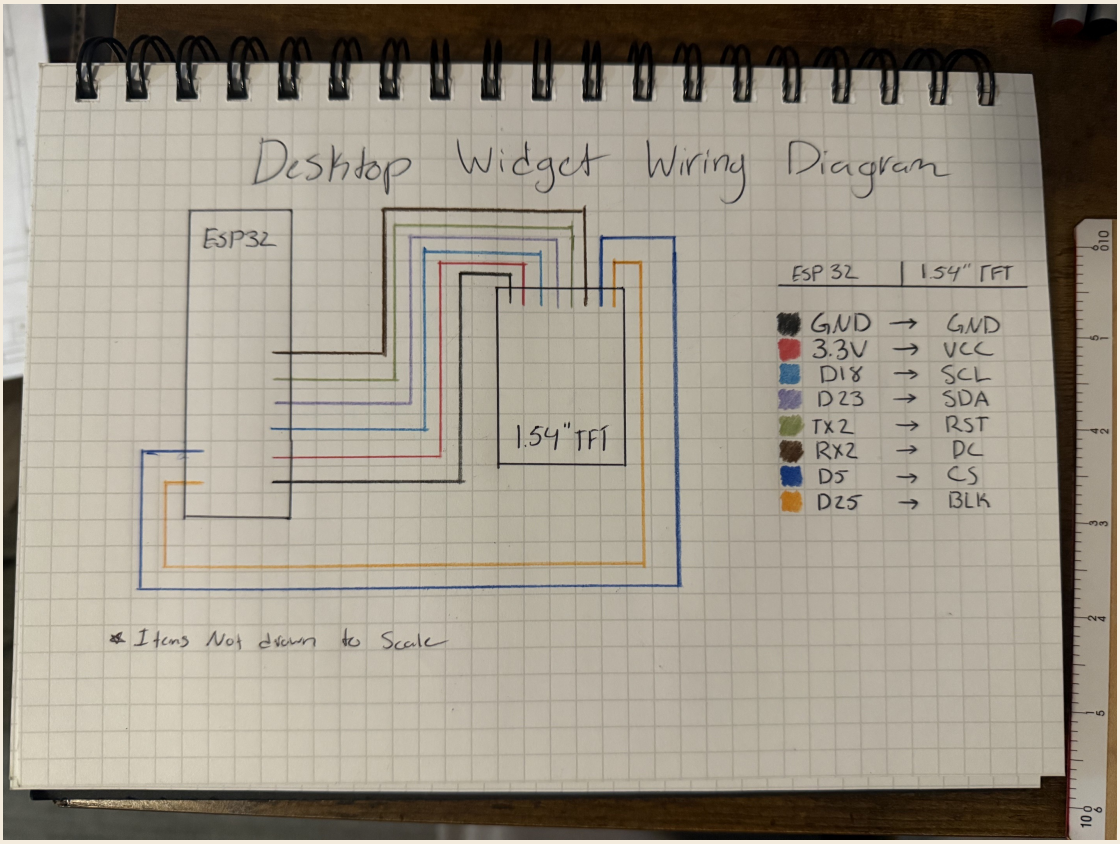


Figure 1. ESP32 to 1.54-inch ST7789 wiring diagram. The drawing is not to scale. Verify every connection by its printed pin label before applying power.

SECTION 02B

Backlight and auto-dimming

Choose one backlight option. The safest simple option is constant brightness. Automatic dimming requires the optional sensor and a suitable backlight control method.

Option A - Constant brightness

Connect BLK to 3.3 V. This keeps the backlight continuously on and avoids drawing backlight current through a GPIO pin. Use this option during the first test even if automatic dimming will be added later.

Option B - Direct PWM control

Connect BLK to GPIO 25 only when the module documentation or a current measurement confirms that BLK is a low-current logic control input. Do not assume this from the pin name alone. Set GPIO 25 as a PWM output in the firmware.

Option C - Transistor-controlled PWM

Use a logic-level MOSFET or transistor when BLK directly powers the backlight LEDs. This lets GPIO 25 control brightness without carrying the LED current. Build this option only from the transistor manufacturer's pinout and a circuit verified for the exact display module.

Optional BH1750 light sensor

BH1750 pin	ESP32 pin	Purpose
VCC	3.3 V	Sensor power
GND	GND	Common ground
SCL	GPIO 22	I2C clock
SDA	GPIO 21	I2C data
ADDR	Leave at module default	Normally selects I2C address 0x23

Place the sensor behind a small opening on the front or upper surface of the enclosure. Keep it out of the TFT's direct light and away from deep shadows. A short recessed light tunnel helps it measure the room instead of the screen itself.

SECTION 03

Prepare the Arduino IDE

Complete this section before connecting the assembled circuit. The CP2102 driver may be required before a COM port appears.

1. Install Arduino IDE 2.x
   Download and install the current Arduino IDE from the official Arduino website. Start the IDE once after installation.

2. Install the CP210x USB driver if needed
   Connect the ESP32 with a known data cable. If Windows Device Manager shows a yellow warning symbol or no COM port appears, install the Silicon Labs CP210x VCP driver, reconnect the board, and restart the IDE.

3. Add ESP32 board support
   Open File > Preferences. Add `https://espressif.github.io/arduino-esp32/package_esp32_index.json` to Additional Boards Manager URLs. Then open Tools > Board > Boards Manager, search for ESP32, and install ESP32 by Espressif Systems.

4. Select the board and port
   Choose Tools > Board > ESP32 Arduino > ESP32 Dev Module. Choose the COM port that appears when the board is plugged in. Leave the default flash settings unless the supplied firmware says otherwise.

Required libraries

Library	Install from	Used for
Adafruit GFX Library	Arduino Library Manager	Drawing text, shapes, and display graphics.
Adafruit ST7735 and ST7789 Library	Arduino Library Manager	Driving the 240 x 240 ST7789 TFT.
ArduinoJson	Arduino Library Manager	Parsing weather and location responses.
WiFiManager	Arduino Library Manager	Phone-based Wi-Fi setup portal.
BH1750	Arduino Library Manager, optional	Reading room brightness when the sensor is fitted.

The ESP32 core already supplies WiFi, HTTPClient, time, Preferences/NVS, and other standard ESP32 features. Do not install unrelated libraries with similar names unless the firmware documentation specifically requests them.

SECTION 04

Install and configure the firmware

The complete Arduino sketch will be supplied with this manual as a separate .ino project. Keep every tab or supporting file inside the same sketch folder.

1. Open the supplied sketch
   In Arduino IDE, choose File > Open and select the main `.ino` file. If Arduino offers to move it into a matching folder, allow it to do so.

2. Confirm the pin definitions
   Verify that TFT_SCLK is 18, TFT_MOSI is 23, TFT_RST is 17, TFT_DC is 16, TFT_CS is 5, and TFT_BL is 25 when PWM control is installed. Match the firmware option to the selected backlight wiring method.

3. Review user settings
   Set the preferred temperature unit, display rotation, time format, weather refresh interval, default location behavior, and optional GitHub message URL. Do not place private passwords or access tokens in a public repository.

4. Compile before uploading
   Click Verify. Resolve every missing-library or syntax error before connecting the final circuit. Warnings may be acceptable, but compilation must finish successfully.

5. Upload
   Select ESP32 Dev Module and the correct COM port, then click Upload. If the IDE remains at Connecting, hold the BOOT button, start the upload, and release BOOT when writing begins. Press EN/RESET once after upload if the program does not start automatically.

Expected firmware features

Feature	Expected behavior
Animation	Sisyphus remains near center while rocky terrain scrolls beneath him at about 10 FPS.
Clock	Local time includes seconds and AM/PM after NTP synchronization.
Weather	Current temperature and animated clear, cloud, rain, storm, fog, or snow conditions.
Moon	Phase advances according to the current date and approximate 29.53-day lunar cycle.
Mallard	May land no more than twice per 24 hours and causes a two-minute walking break.
Mileage	Increases at 3.2 mph while walking and saves to NVS approximately every 15 minutes.

SOURCE CODE

Open `SisyphusDeskDisplay.ino` in Arduino IDE. Keep it inside a folder named `SisyphusDeskDisplay` so the Arduino project name and main sketch filename match.

SECTION 05

First start and Wi-Fi setup

Perform the first start with the circuit still visible on the breadboard. Keep a phone or laptop nearby.

1. Apply power
   Connect the ESP32 to the computer or a stable USB supply. The screen should illuminate and show a startup or connection screen within several seconds.
2. Join the setup network
   If the device has no saved Wi-Fi network, open the Wi-Fi settings on a phone or laptop and join `DeskDisplay-Setup`. A setup page should open automatically. If it does not, open a browser and visit `192.168.4.1`.
3. Select the home network
   Choose the intended 2.4 GHz Wi-Fi network, enter its password carefully, and save. The setup network will close while the ESP32 restarts and joins the selected network.
4. Wait for synchronization
   Allow up to a few minutes for location, weather, time, and moon data to settle. Confirm that the displayed time zone and temperature are reasonable for the installation location.
5. Verify offline behavior
   After a successful update, briefly disable the Wi-Fi access point. The animation and saved mileage should continue. Live time and weather updates will resume when the network returns.

Changing networks later

Use the firmware's Wi-Fi reset action to erase only the stored network credentials, then reconnect to DeskDisplay-Setup. The exact button or reset sequence depends on the final firmware and will be inserted here when the code is finalized. Do not erase the entire flash unless instructed, because doing so may also erase saved mileage.

SECTION 06

Install the enclosure

Only transfer the circuit after it runs correctly for at least one continuous hour on the breadboard.

Recommended physical arrangement

Mount the TFT close to the front opening with a small bezel covering the unused edges. Position the ESP32 behind or below it with the USB-C connector accessible from the rear. Leave air space around the voltage regulator and ESP32 antenna. Do not place metal fasteners, copper tape, or a large wire bundle directly over the antenna end of the development board.

Assembly sequence

1. Disconnect USB power. 2. Mount the display without pressing on the LCD surface. 3. Secure the ESP32 on insulated standoffs. 4. Route short signal wires away from the USB opening. 5. Place the BH1750 behind its light opening, if fitted. 6. Add strain relief to the USB cable path. 7. Inspect for trapped or pinched wires. 8. Reconnect power before closing the enclosure for the final test.

Do not use hot glue directly on the ESP32 antenna, USB connector, display flex cable, or exposed components that become warm. Double-sided electronics tape, printed brackets, or small nonconductive standoffs make later repairs easier.

Screen orientation

If the image is upside down after mounting, change the ST7789 rotation value in the firmware and upload again. Rotation values normally range from 0 through 3. Test all four if the correct value was not documented.

SECTION 07

Commissioning checklist

Complete every check before giving the finished display to another person or installing it far away.

Electrical

[ ] VCC measures approximately 3.3 V relative to GND.

[ ] No ESP32 GPIO pin is connected to 5 V.

[ ] The display, ESP32, and optional sensor share a common ground.

[ ] The backlight circuit matches the selected constant, direct-control, or transistor option.

[ ] No bare conductor can touch a neighboring pin or the enclosure.

Display and animation

[ ] The screen starts without cycling through solid colors indefinitely.

[ ] The image is upright and fills the 240 x 240 area.

[ ] Animation is smooth and the screen does not flash between frames.

[ ] The bottom Camus-inspired line is visible and not clipped.

[ ] The total mileage is readable and does not reset after ordinary power loss.

Network and live data

[ ] DeskDisplay-Setup appears when no network is saved.

[ ] The device reconnects automatically after the router or device restarts.

[ ] Local time, seconds, and AM/PM are correct.

[ ] Temperature and weather agree reasonably with the installation location.

[ ] The moon phase is plausible for the current date.

Long-duration test

[ ] The device has operated for at least 8 hours without restarting or freezing.

[ ] The enclosure remains only mildly warm and has no hot spots.

[ ] Auto-dimming responds smoothly without rapid brightness flicker, if installed.

[ ] A power cycle restores normal operation and preserves accumulated mileage.

Record the firmware version, test date, Wi-Fi reset procedure, and starting mileage on the service record at the end of this manual.

SECTION 08

Troubleshooting: power and display

Always disconnect USB before changing wires. Change one thing at a time and retest after each change.

Symptom	Likely cause	Corrective action
No COM port	Charge-only cable, missing CP210x driver, bad USB port	Try a known data cable and another USB port. Check Device Manager. Install/reinstall the Silicon Labs CP210x VCP driver.
Yellow warning icon	Driver failed or device identification problem	Remove the failed device in Device Manager, reconnect it, install the correct CP210x driver, and restart Windows.
ESP32 will not upload	Wrong port/board, bootloader not entered	Select ESP32 Dev Module and the correct port. Hold BOOT while upload begins, then release when writing starts.
Backlight is off	BLK not powered, wrong polarity, unsafe direct control	Temporarily connect BLK to 3.3 V after confirming the module pinout. Recheck GND and VCC.
Lit but blank	Wrong CS/DC/RST pins or failed initialization	Verify GPIO 5, 16, and 17 by printed labels. Confirm ST7789 240 x 240 initialization in code.
Solid color test works only	Display hardware is alive; main sketch setup/rendering is failing	Open Serial Monitor, inspect startup errors, confirm libraries, and test the final sketch with weather features temporarily disabled.
Wrong colors	RGB/BGR order mismatch or incorrect inversion	Adjust ST7789 color order or inversion option in the firmware.
Image upside down	Rotation does not match mounting	Change the display rotation value from 0 to 1, 2, or 3 and upload again.
Flashing screen	Full-screen clearing or insufficient buffered drawing	Use the supplied strip-buffer renderer. Do not clear the entire screen between every animation frame.
Random lines/noise	Loose SPI wiring, wires too long, weak power	Shorten and reseat wires, verify common ground, and try a stable USB supply.

Quick isolation test

Upload a simple red-green-blue-white-black screen test. If each full-screen color appears correctly, the display, SPI bus, and power connections are fundamentally working. Return to the main sketch and diagnose initialization, memory, or rendering logic rather than replacing the screen.

SECTION 08B

Troubleshooting: Wi-Fi and live data

Use Serial Monitor at the baud rate specified by the supplied firmware. Startup messages are the fastest way to separate a network problem from a display problem.

Symptom	Likely cause	Corrective action
Setup network does not appear	Credentials are already saved or WiFiManager did not start	Use the documented Wi-Fi reset action. If unavailable, upload a credential-clear utility only after preserving mileage data.
Portal opens but cannot connect	Wrong password, unsupported 5 GHz-only network, captive portal	Verify the password and use a 2.4 GHz network. Hotel/school captive portals may not work without custom support.
Time is incorrect	Wrong time zone or NTP unavailable	Confirm internet access, time-zone configuration, and NTP server response. Restart after correcting settings.
Weather is wrong	Approximate IP location is inaccurate	Use a manually configured latitude and longitude in the final firmware rather than network-assisted location.
Weather never updates	API request blocked, URL changed, JSON parsing error	Read the HTTP status in Serial Monitor. Verify Open-Meteo request fields and ArduinoJson capacity.
Message will not change	GitHub URL is private, incorrect, cached, or not raw content	Use a public raw-file URL or approved authenticated method. Confirm the device receives HTTP 200 and plain text.
Frequent restarts	Weak power supply, memory pressure, watchdog timeout	Use a stable 1 A supply, shorten blocking network operations, and inspect free heap messages.
Mileage resets	Preferences/NVS save is missing, erased, or using a changed key	Restore the original storage namespace/key and verify that the 15-minute save routine succeeds.
Auto-dimming flickers	Sensor sees the TFT or raw readings are unsmoothed	Move/shield the sensor and average readings. Add hysteresis and change brightness gradually.

Remote installations

Before the display is sent to a distant location, confirm that the recipient can reach the USB connector and knows how to reopen the Wi-Fi setup portal. Do not depend on local-network Arduino OTA for internet-distance updates. Any true remote update system must use an authenticated, encrypted update source and a recovery plan for interrupted firmware downloads.

Never publish Wi-Fi passwords, private tokens, or personal location coordinates inside a public GitHub repository.

SECTION 09

Maintenance and care

The display is intended for indoor desktop use. It contains exposed low-voltage electronics and is not weatherproof.

Normal operation

Leave the device powered from a reliable USB adapter. Clean the screen with a dry microfiber cloth and light pressure. Do not spray cleaner into the enclosure. Keep the ventilation openings clear and avoid direct sunlight, heaters, high humidity, and locations where liquid can spill onto the unit.

Firmware changes

Keep a dated backup of every firmware version that is installed. Before changing storage keys or erasing flash, record the current mileage. After an update, verify display rotation, weather, time zone, moon phase, mallard limits, 3.2 mph mileage rate, pause behavior, milestone thresholds, and NVS saving.

Boulder milestone reference

Mileage	Boulder appearance	Celebration
10	Scratched stone	No major celebration
100	Smoother stone	No major celebration
500	Bronze veins	No major celebration
1,000	Bronze	8-second milestone sequence
5,000	Polished bronze	Milestone sequence
10,000	Silver	Milestone sequence
25,000	Polished silver	Flashes, sparkles, circling mallard
50,000	Glowing silver	Milestone sequence
75,000	Gold-veined silver	Flashes, sparkles, circling mallard
100,000	Permanent gold	Final material transformation; journey continues

Mileage should increase only while Sisyphus is walking. The two-minute mallard rest pauses distance accumulation. At 3.2 mph, one real hour of active walking adds approximately 3.2 simulated miles.

SECTION 10

Builder handoff

Complete this page before giving the display to its owner. Keep a second copy with the project files.

Files to include

[ ] Verified Arduino project folder containing the .ino file and every supporting tab.

[ ] Library list and known-working version numbers.

[ ] Final wiring diagram matching the actual build.

[ ] Enclosure photographs showing internal wire routing.

[ ] Instructions for reopening DeskDisplay-Setup after moving to a new network.

[ ] Recovery instructions for a failed firmware upload.

[ ] Backup of the current mileage and firmware configuration.

Owner quick guide

To move the display to a new Wi-Fi network, activate the documented Wi-Fi reset action, join DeskDisplay-Setup from a phone, and select the new 2.4 GHz network. Normal USB power loss will not erase mileage. If the screen is blank, disconnect power, wait ten seconds, reconnect it, and check that the backlight turns on. Do not open the enclosure or move internal wires while power is connected.

Service record

Item	Recorded value
Build date	
Builder	
Firmware filename/version	
ESP32 board/USB interface	ESP32-WROOM-32 / CP2102
Display	1.54 inch ST7789, 240 x 240
Display rotation	
Backlight wiring option	
BH1750 fitted	Yes / No
Starting mileage	
Long-duration test completed	
Wi-Fi reset procedure	